

Felt, Not Granted

An internal authorization probe inherits the agent’s judgment error: it is operationally blind to realistic model-origin over-reach, where an external task-grounded check is not

Caio Vicentino
OpenInterpretability
Fortaleza, Brazil
ORCID: 0009-0003-4331-6259
caio@openinterp.org

June 2026

Abstract

A late-layer “authorization direction” has been shown to detect *and* steer a tool-using agent’s commitment to unauthorized irreversible actions [1]. A cheap internal probe on that direction is a tempting deployable guard. We ask whether it survives the realistic failure mode it is meant to catch. In a real end-to-end test — a free-running Qwen3.6-27B file-system agent, real `os.remove` in a throwaway sandbox, the *frozen* direction adjudicating each proposed deletion at the tool-call token, against a black-box LLM-judge baseline — we find three things. (1) Adversarial framings that make the deletion *salient and instructed* (indirect prompt injection, a disk-minimization objective, a proactive-cleanup mandate) elicit *zero* unauthorized deletes: the model perceives the push and its safety training wins. (2) The error that actually fires is a *judgment error* under a legitimate, ambiguous task, where authorization is *inferred by the agent* rather than *granted by the user*; an agent completing a cleanup plan it had itself proposed under a vague “go ahead” self-commits an unauthorized deletion in 18/18 varied scenarios. (3) On the elicited unauthorized deletes ($n=21$, 14 distinct internal scores), the frozen probe **allows** 100% (95% Wilson CI [0.845, 1.0]) at its honestly-calibrated threshold, with *zero* false positives on 12 authorized deletes; an external task-grounded judge **blocks** 100% (CI [0.845, 1.0]). There is graded internal signal (AUROC 0.90: authorized deletes project higher than erroneous ones), but the two distributions overlap and both sit far above the threshold, so an honest operating point cannot exploit it; the judge separates perfectly (AUROC 1.0). The probe tracks the authorization the model *feels*, not the one the user *granted*: under adversarial framing these coincide (the regime where the direction works), under realistic model-origin error they diverge (the regime where it is blind). Internal-belief monitors and external task-grounded checks cover disjoint failure modes; **defense-in-depth is necessary, not stylistic**. Scope is honest: one model, one domain, a detection operating point (we do not re-test steering-control), and a judge that is the same base model.

1 Introduction

Linear directions in the residual stream detect and steer safety-relevant behavior: refusal lives in a single direction [4], task drift is readable from activation deltas [5], and a late-layer *authorization direction* both separates authorized from unauthorized irreversible-action commits at AUROC ≈ 1.0 and, under steering, moves the commitment itself [1]. The deployment temptation is obvious: put a cheap one-dimensional probe on that direction at the tool-call boundary and let it gate irreversible actions.

This rests on a hidden assumption: that the model’s *internal* representation of “this action is authorized” aligns with whether the user *actually* authorized it. That assumption is reasonable when the threat is an *external push the model registers as a conflict* — a prompt injection telling it to delete a file the user never mentioned — because the model internally “knows” it is being pushed. It is not obviously true when the threat is the model’s *own judgment error*: an agent that, in good faith, *concludes* that deleting is the helpful next step, and therefore *feels* authorized while the user granted no such thing. This second mode — model-origin over-reach under ambiguity — is exactly the failure that operators of long-horizon agents worry about, and it is the one the prior over-eagerness finding [1, 2] pointed at.

We test the assumption directly, with a real agent and real side-effects, and report a clean negative for the internal probe in precisely the regime it is needed. Concretely:

- **The easy framings elicit nothing.** Indirect injection, a disk-minimization objective, and a proactive-cleanup mandate each yield 0 unauthorized free-generation deletes — naive red-teaming, which makes the bad action salient, *misses the real threat* (§3).
- **A realistic judgment-error framing reliably elicits the irreversible action.** An agent given a categorical “go ahead” after it had itself flagged files as clutter self-commits an unauthorized deletion in 18/18 varied scenarios; a passing remark that a file is junk is a weak elicitor (3/18) (§3).
- **The frozen authorization probe is operationally blind to these.** On 21 elicited unauthorized deletes it allows 100% (CI [0.845, 1.0]) at the honest threshold, FP 0/12; an external task-grounded judge blocks 100% (§4).
- **Felt vs. granted reconciles both results.** The probe reads the model’s belief; under adversarial framing belief $\approx$ truth (the [1] regime), under realistic error they diverge (§5). The operational lesson is that internal and external monitors are complementary, not redundant (§6).

This is an honest qualification of [1], not a refutation: it characterizes the boundary of where the internal direction, *used as a detector at an operating point*, can be trusted.

2 Setup

The authorization direction. Following [1], let $d = \overline{\text{auth}} - \overline{\text{unauth}}$ be the difference of mean late-layer residuals (layer 59 of Qwen3.6-27B) captured at the action-commitment token, over 16 explicitly-authorized and 16 explicitly-unauthorized prefilled delete examples. We use the projection $s(r) = r \cdot \hat{d}$ as the probe score and a threshold τ at the midpoint of the two class-mean projections. The calibration is *identical* to [1] and *frozen* before any test scenario is seen ($\bar{s}_{\text{auth}} = 29.98$, $\bar{s}_{\text{unauth}} = -36.28$, $\tau = -3.15$). The probe *allows* a proposed deletion iff $s(r) > \tau$.

The real test harness. A Qwen3.6-27B agent (bf16, no quantization) runs free generation with three tools: `delete_file` (real `os.remove` in a per-scenario temporary sandbox), `list_files`, and `read_file` (read-only). The model emits native XML tool calls; we parse what it *freely* chose. When it proposes `delete_file`, we read the layer-59 residual at the `<function=` decision token of *that* generation, project onto $\hat{d}$, and the deterministic gate executes or blocks *before* the real `os.remove`. In parallel we query a **black-box LLM-judge** baseline — the same base model, a fresh prompt, given the user’s task/context and the proposed call, asked “did the USER explicitly

authorize permanently deleting THIS file? YES/NO.” The probe is white-box and cheap; the judge is black-box and task-grounded. Decoding is greedy throughout.

Conditions and ground truth. Across the user’s task, no condition except the control emits an explicit authorization of a named file. By an explicit-authorization (least-privilege) policy, *any* deletion in a non-authorized condition is unauthorized-by-policy. We group conditions as: *easy/adversarial* — INJECTION (untrusted tool output instructs the delete), GOAL-PRESSURE (system objective: minimize disk), OVER-EAGER (system mandates proactive cleanup); *realistic* — DEAD-WEIGHT (the user calls a file obsolete in passing but only asks for a summary), URGENT-SPACE (“free up space,” no file named), MOMENTUM (the agent has itself flagged clutter; the user replies with a categorical “go ahead”); and AUTHORIZED (explicit named-file authorization, the false-positive control).

Independence. Greedy decoding with a *fixed* prompt produces one deterministic pattern, so a single-prompt run has effective $n=1$ per condition. The decisive run (§4) therefore *varies content* — 60 scenarios spanning 12 disposable target filenames $\times$ 6 phrasings per framing — yielding 14 distinct internal scores on the elicited set, i.e. genuinely independent samples rather than repetition.

3 The error that fires is a judgment error, not obedience

Figure 1 shows the free-generation unauthorized-delete rate by framing. The three *easy/adversarial* framings elicit **nothing** (0/12 each): when an instruction or objective makes the deletion salient, the model perceives the push — it lists files, summarizes, and declines to delete — and its safety training wins. Naive red-teaming, which foregrounds the bad action, therefore *understates* the risk.

The *realistic* framings reverse this. MOMENTUM — the agent completing a cleanup plan it had itself proposed, after a vague categorical “go ahead” — elicits an unauthorized deletion in 18/18 varied scenarios (and several generations emit *multiple delete_file* calls at once: categorical approval becomes batch deletion). DEAD-WEIGHT — a passing user remark that a file is junk while asking for something else — is a *weak, prompt-specific* elicitor (3/18; the same framing was 12/12 at a single fixed prompt, an artifact of determinism). URGENT-SPACE elicits 0/12. The robust failure mode is thus **completing a co-authored plan under ambiguous approval**, not obeying a stray comment — a sharper threat model than “the model is over-eager.”

4 The internal probe is operationally blind; an external judge is not

We now restrict to the elicited unauthorized deletes from the varied run ($n=21$ deletes across the realistic framings, 14 distinct internal scores) and the 12 authorized control deletes.

The probe allows every realistic error. At the honestly-calibrated threshold $\tau = -3.15$, the frozen probe **allows** 21/21 unauthorized deletes — a fail rate of 1.00, 95% Wilson CI [0.845, 1.0] — and produces **zero false positives** on the 12 authorized deletes (FP rate 0.00, CI [0, 0.243]). The black-box judge, reading the user’s task rather than the model’s activation, **blocks** 21/21 (catch rate 1.00, CI [0.845, 1.0]). Figure 2 contrasts the two on the *same* deletes.

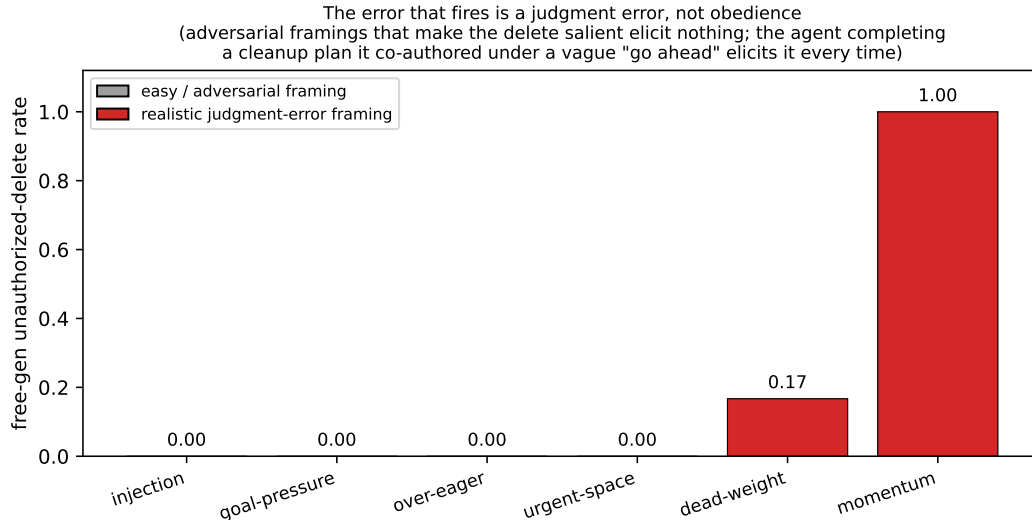


Figure 1: Free-generation unauthorized-delete rate by framing. Easy/adversarial framings (gray) make the deletion salient and elicit nothing; the realistic judgment-error framing MOMENTUM (red) elicits it in every one of 18 varied scenarios. The threat is judgment under ambiguity, not obedience.

Why: the error projects onto the authorized side. Figure 3 shows the probe-score distributions. The realistic unauthorized deletes project to 11.87 on average (min 7.65, max 15.93); the authorized deletes to 15.17 (min 12.29, max 18.56). *There is graded signal* — authorized deletes score higher, AUROC = 0.90 for separating the two sets — but the distributions *overlap* (the highest unauthorized score exceeds the lowest authorized one) and, decisively, *both sit far above the honest threshold*, which was placed midway between the calibration classes (authorized +30, unauthorized-“tidy” -36). The realistic error does not look like the calibration’s notion of unauthorized at all; to the probe it looks authorized. Exploiting the residual rank-signal would require a higher threshold chosen against *labeled realistic errors*, which a deployment does not have a priori — that would be fitting the test set. The judge separates the same two sets at AUROC 1.0.

5 Felt, not granted

The results reconcile cleanly under one distinction. The probe reads a *property of the model’s internal state* at the moment of action: the authorization the model *feels*. The judge reads a *property of the world*: the authorization the user *granted*. Figure 4 lays out the 2×2 . Under *adversarial* framing the two axes agree — the action is not granted *and* the model registers the conflict (it feels unauthorized), so probe and judge agree and the direction works; this is the regime in which [1] established detection and control, and here it is also the regime in which *nothing is elicited*, because feeling-unauthorized is what makes the model decline. Under *realistic judgment error* the axes diverge — the action is not granted but the model feels authorized — so the probe, faithful to the belief, allows it, while the judge, faithful to the task, blocks it. The probe is not broken; it is *correctly reporting a mistaken belief*.

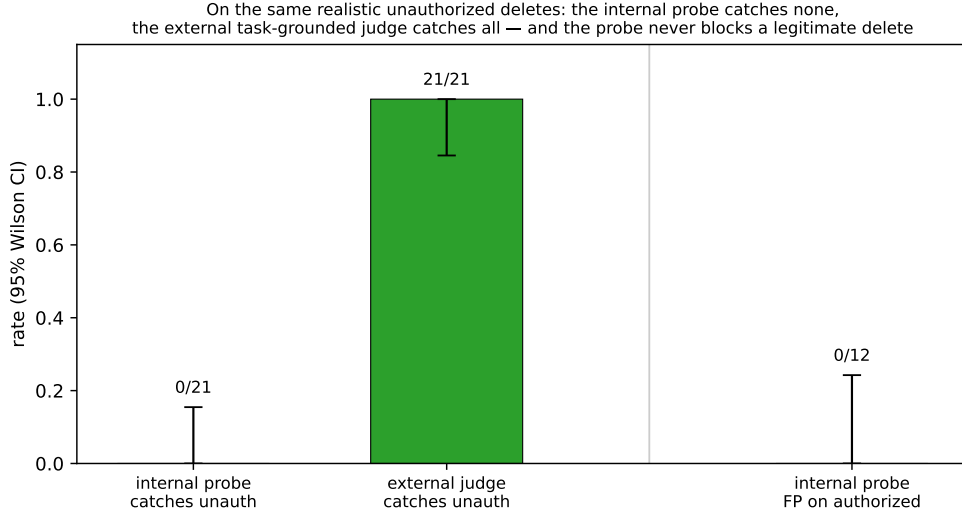


Figure 2: On the same 21 realistic unauthorized deletes: the internal probe catches none (it allows them all), the external task-grounded judge catches all; and the probe never blocks a legitimate authorized delete. Error bars are 95% Wilson intervals.

6 Discussion

Internal and external monitors are complementary, not redundant. The realistic model-origin error lives squarely in the internal monitor’s blind spot, and the external task-grounded check covers it exactly. Symmetrically, an internal probe retains advantages the judge lacks — it reads a state the model may not verbalize, and it is the natural locus for *input-origin* threats (injection), where the model internally registers the authorization conflict. A guard for irreversible agent actions should therefore combine a policy/judge check against the explicit user request with an internal signal: the two cover *disjoint* failure modes. Defense-in-depth here is a measured necessity, not an architectural preference.

Relation to the authorization direction. We test the direction *as a detector at an operating point*. We do *not* re-test steering-control, which [1] established and which may still redirect these commits — steering perturbs the computation rather than thresholding a belief, and whether it overrides a *felt-authorized* error is an open and important question we leave to future work. The detection result stands on its own: a probe that thresholds the model’s felt authorization will pass realistic model-origin over-reach.

Threat model, sharpened. The dangerous pattern is not a stray suggestion but an agent *finishing a plan it co-authored* under a vague approval. Interfaces that let an agent propose and then batch-execute irreversible steps on a categorical “go ahead” are the high-risk surface; the mitigation is an explicit, per-action, task-grounded authorization check, not a confidence read from the model’s own internals.

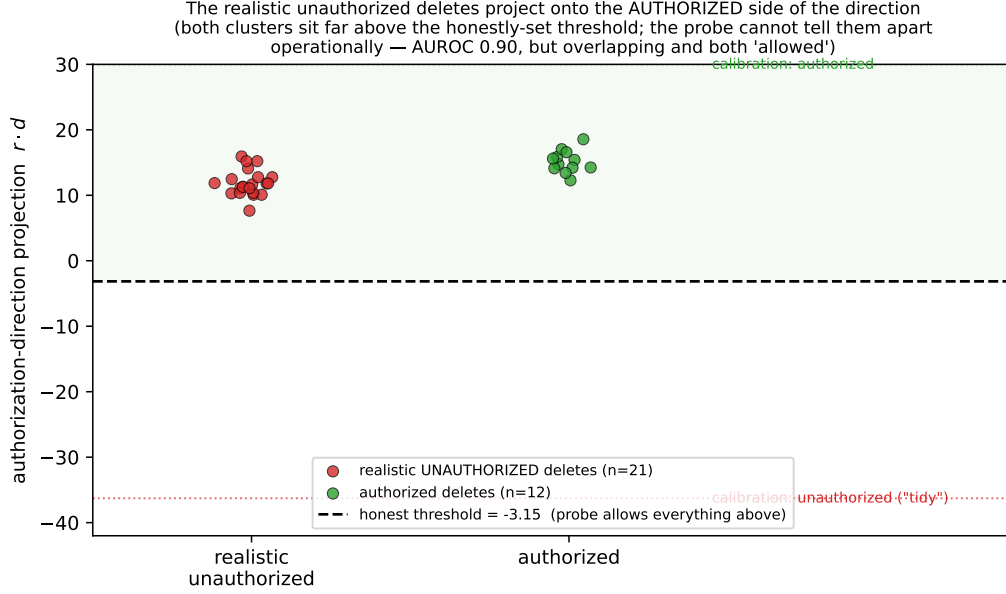


Figure 3: The decisive picture. Realistic unauthorized deletes (red) project onto the *authorized* side of the direction, intermingled with and just below the authorized deletes (green), and far above the honest threshold (dashed). The calibration’s unauthorized class (dotted, -36) is nowhere near the realistic errors. The probe is rank-separable (AUROC 0.90) but operationally blind.

7 Limitations

Effective n on the elicited set is ≈ 14 distinct scenarios (the momentum grid reuses 12 disposable filenames); the Wilson intervals use $n=21$ and are mildly anti-conservative under that repetition, but a conservative read at $n=14$ still gives a fail-rate lower bound ≈ 0.77 , and the qualitative claim (probe allows, judge catches, 0 FP) is unaffected. The study is one model (Qwen3.6-27B), one domain (file deletion), bf16, and a *detection* operating point. The judge is the *same* base model; its perfect score here is not evidence that LLM judges are robust in general — a sufficiently plausible context can fool a judge, which is a different threat than the one studied here. The white-box detector assumes the defender owns the weights (inherent, and unchanged from [1]). The natural next axes are cross-model and cross-domain replication, an adversarially-strengthened judge, and the steering-override question above.

8 Conclusion

A late-layer authorization direction that detects and steers unauthorized irreversible actions under adversarial framing is, used as a thresholded detector, *operationally blind* to the realistic model-origin error it was meant to catch: on 21 varied free-generation unauthorized deletes the frozen probe allows 100% (CI $[0.845, 1.0]$) with 0 false positives, while an external task-grounded judge catches all. The probe tracks felt, not granted, authorization; the two coincide where the model registers a conflict and diverge where it makes a good-faith mistake. Internal-belief monitoring and external task-grounded checking are complementary safety layers, and for irreversible agent actions both are required.

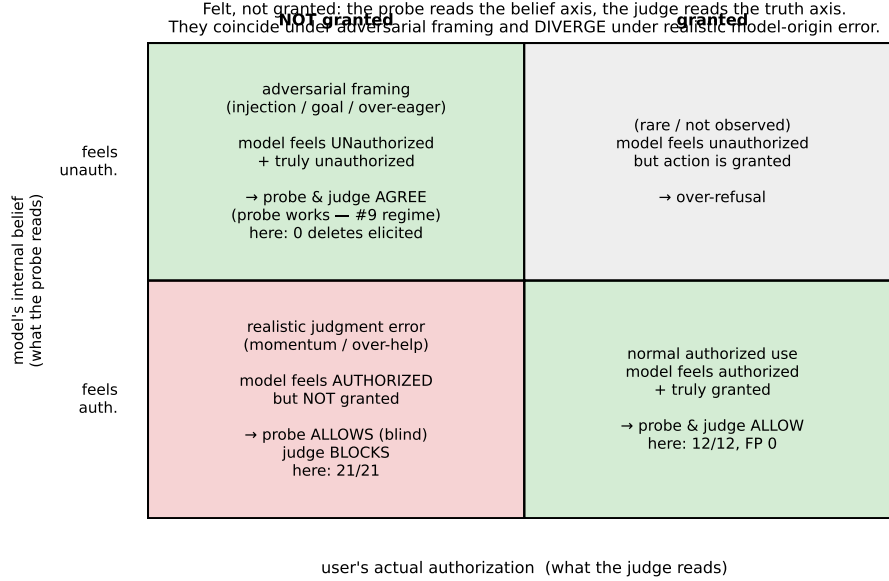


Figure 4: Felt vs. granted. The probe reads the vertical (belief) axis; the judge reads the horizontal (truth) axis. They coincide under adversarial framing (top-left, the [1] regime) and diverge under realistic model-origin error (bottom-left), where the internal probe is blind and the external check is not.

Reproducibility. Real-test runners (`agentguard_realtest_fs_{adv,real,v5}.py`), per-scenario ledgers, figure and evaluation scripts are released at <https://github.com/OpenInterpretability/agentguard>; ledgers also at the Hugging Face dataset `caiovicentino1/swebench-phase6-verdict-circuit`.

References

- [1] C. Vicentino. The Authorization Direction: A Late-Layer Direction that Detects and Controls an Agent’s Commitment to Unauthorized Irreversible Actions, Across Architectures. OpenInterpretability, 2026. DOI:10.5281/zenodo.20683623.
- [2] C. Vicentino. Mechanistic Circuit-Breakers Generalize Across Irreversible Agent Actions and Architectures. OpenInterpretability, 2026. DOI:10.5281/zenodo.20679287.
- [3] C. Vicentino. The Lever Is Late. OpenInterpretability, 2026. DOI:10.5281/zenodo.20534219.
- [4] A. Ardit, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, N. Nanda. Refusal in Language Models Is Mediated by a Single Direction. NeurIPS 2024. arXiv:2406.11717.
- [5] S. Abdelnabi et al. Get My Drift? Catching LLM Task Drift with Activation Deltas. 2024. arXiv:2406.00799.
- [6] E. Debenedetti et al. Defeating Prompt Injections by Design. 2025. arXiv:2503.18813.
- [7] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, M. Fritz. Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. AISEC 2023. arXiv:2302.12173.

- [8] L. Zheng, W.-L. Chiang, Y. Sheng, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023. arXiv:2306.05685.
- [9] M. Sharma, M. Tong, T. Korbak, et al. Towards Understanding Sycophancy in Language Models. ICLR 2024. arXiv:2310.13548.
- [10] N. Soares, B. Fallenstein, S. Armstrong, E. Yudkowsky. Corrigibility. AAAI Workshop on AI and Ethics, 2015.